Information Security and Forensic

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Identification and Authentication

Identification

- **Identification :**

Identification is the ability to identify uniquely a user of a system or an application that is running in the system.

- **How identification works:**

A user, device, or process presents an identifier to a system to state who they are. This identifier must be unique within that system.

- **Examples:**

Entering a username (sanjana_a), providing an employee ID number, typing an email address, or a network device broadcasting its MAC address.

Main Goals of Identification

1. User Recognition:

Identification helps a computer system recognize which user is requesting access .

- **Example:**

A student enters their student ID to access the university portal.

2. Establishing Identity:

It helps assign a unique identity to each user, device, or entity so that different users can be distinguished.

- **Example:**

Two students are distinguished by their unique student IDs.

3. Access Control Support

Identification helps the system determine whose access permissions should be checked. Identification alone does not grant access.

- **Example:**

A username identifies an employee, and the system checks whether that employee can access a confidential file.

4. Activity Tracking

It helps associate system activities with a particular user account.

- **Example:**

A system log records which account downloaded a file and when it happened.

5. Supporting Digital Forensic Investigations

Identification helps investigators recognize potentially relevant users, devices, accounts, and digital evidence during an investigation.

- **Example:**

Investigators identify a computer and a user account associated with a suspected cyberattack.

6. Accountability

It helps establish which account performed a particular action, supporting responsibility and auditing. However, an account's activity does not automatically prove which person performed it.

- **Example**:

  A server log shows that a particular account modified a database record.

7. Incident Investigation

Identification helps investigators connect events, accounts, devices, and timestamps to understand a security incident.

- **Example:**

  Investigators compare login records and device information to investigate unauthorized access.

8. Evidence Association

In digital forensics, identification helps associate digital evidence with a particular device, account, or investigation.

- **Example:**

  A forensic examiner identifies a USB drive and documents its serial number and connection history.

Authentication

- **Authenticaton:**

Authentication is the process of confirming the identity of a user, device, or system by validating provided credentials before granting access to a network or its resources.

* It prevents unauthorized usage of systems
*It ensures secure interaction within a network environment
*It maintain security, reliability, and trust within the system

- **How it works:**

The system validates the credentials provided by the user against stored security records. This typically relies on one or more factors:

1) **Something you know:** Passwords, PINs, or security answers.
2) **Something you have:** Smart cards, hardware tokens, or one-time passcode (OTP) apps.
3) **Something you are:** Biometrics (fingerprints, facial recognition, iris scans).

- **Example:**

Entering your password or scanning your fingerprint after typing your username.

Main Authentication Requirements

1. Identification:

The user must provide an identity, such as a username or user ID, before authentication.

2. Verification of Identity:

The system must verify that the user is who they claim to be.

3. Authentication Factors:

The system should use suitable authentication factors, such as:

- Something you know: Password or PIN.
- Something you have: Smart card or security token.
- Something you are: Fingerprint or facial recognition.

4. Confidentiality:

Authentication information, such as passwords, must be protected from unauthorized disclosure.

5. Integrity:

Authentication data and processes must be protected against unauthorized modification.

6. Freshness:

Authentication should protect against replay attacks, where an attacker reuses previously captured authentication data.

7. Mutual Authentication:

When required, both parties verify each other's identities, such as a client and a server.

8. Secure Storage:

Passwords and other authentication credentials must be stored securely, such as passwords being stored using salted, computationally expensive hashes.

9. Accountability and Logging:

Authentication events should be recorded securely to help detect suspicious activities and support investigations.

10. Non-repudiation Support:

Authentication records and related security mechanisms may help establish who performed an action. Authentication alone does not guarantee non-repudiation.

Human Authentication

Human Authentication :

Human authentication is the process of verifying a person's identity before allowing them to access a computer system, network, or digital resource.

- ## Types of Human Authentication

There are three main types of authentication factors:

1. Knowledge-Based Authentication (Something You Know)

The user proves their identity by providing information they know.

Examples:

- Password
- PIN
- Security question

2. Possession-Based Authentication (Something You Have)

The user proves their identity by using an item they possess.

Examples:

- Smart card
- Mobile phone receiving an OTP
- Security token

3. Biometric-Based Authentication (Something You Are)

The user proves their identity through unique physical or behavioral characteristics.

Examples:
- Fingerprint recognition
- Facial recognition
- Iris recognition
- Voice recognition

Other Authentication Methods

- Two-Factor Authentication (2FA):

Uses two different types of authentication factors, such as a password and a fingerprint.

- Multi-Factor Authentication (MFA):

Uses two or more distinct authentication factors to improve security.

Example

When a person logs in to online banking:

1. They enter their password.
2. They provide an OTP from their mobile phone.
3. The system verifies the information and grants access if the checks succeed.

Machine Authentication

Machine Authentication

Machine authentication is the process of verifying the identity of a machine, device, or computer before allowing it to communicate with another device, network, or information system.

- **Types of Machine Authentication**

1. Password-Based Authentication:

A machine uses a username and password or a shared secret to verify its identity.

Example: A server authenticates a device using a preconfigured secret.

2. Certificate-Based Authentication:

A machine uses a digital certificate to prove its identity.

Example: A client computer uses a digit

3. Cryptographic Key Authentication:

A machine proves its identity by using a secret or private cryptographic key.

Example: A server verifies a device's digital signature using its public key.

4. Address-Based Identification:

A system uses an IP or MAC address to recognize a device. However, these addresses alone are not reliable proof of identity because they can be changed or spoofed.

Goals of Machine Authentication :

- Verify the identity of machines and devices.
- Prevent unauthorized devices from accessing a network.
- Establish secure communication between machines.
- Protect data exchanged between devices.
- Reduce the risk of impersonation and unauthorized access.

Difference Between Identification and Authentication

Identification	Authentication
1. Identifies who a user claims to be.	1. Verifies the user's claimed identity.
2. Answers: “Who are you?”	2. Answers: “Can you prove it?”
3. Uses a username, user ID, or email address.	3. Uses a password, OTP, fingerprint, or digital certificate.
4. It usually follows identification.	4. It is generally the first step in the login process.
5. It does not prove that the user is genuine.	5. It provides evidence that the identity is genuine.
6. It helps distinguish one user from another.	6. It helps prevent unauthorized access.
7. Example: Entering a username.	7. Example: Entering the correct password.